scientific reports

Check for updates

OPEN

Sparse identification of nonlinear dynamics applied to the acoustic levitation of acoustically large objects

Mehdi Akbarzadeh[1]✉, Benjamin Halkon[1] & Sebastian Oberst[1]

Many studies on acoustic radiation forces, especially those applied to acoustic levitation, focus on characterizing the behaviour of acoustic fields. However, the dynamic response of the levitated objects, particularly those larger than the wavelength limit, remains relatively underexplored. Here, we look to bridge this gap by deriving nonlinear equations of motion for a spherical object trapped under acoustic radiation forces while subject to external excitation. For such a contemporary scenario, the otherwise elemental Gorkov formulation fails to provide accurate results. Using Sparse Identification of Nonlinear Dynamical Systems (SINDy), first, we derive the corresponding nonlinear equation of motion from analytical time series data obtained through the Gorkov formulation and external excitation for acoustically small objects. This approach recovers the governing equation with less than 0.05% error in coefficient values when compared to the analytical solution. Second, we conduct experiments with the TinyLev levitator with external excitation applied via an external actuator to generate the required time series for an acoustically large object. SINDy is applied to reconstruct governing equations from experimental data, allowing for the study of how excitation amplitude affects acoustically large objects. All obtained coefficients change with excitation amplitude, and the coefficients in the dynamic equation of motion should not be treated as constants. Strong velocity-dependent terms emerged, indicating a complex relationship between viscosity and object response, which classical models do not predict. The bifurcation diagram obtained using the SINDy-derived equation of motion shows closer agreement with that obtained experimentally. These results demonstrate that SINDy can recover equations consistent with Gorkov's formulation and extend beyond it, providing a pathway to derive analytical expressions directly from data for levitating and manipulating objects beyond the Rayleigh limit.

Keywords Acoustic radiation force, Sparse identification of nonlinear dynamical systems, Acoustic levitation

Acoustic radiation force is that which is experienced by an object when placed in a sound field, resulting from momentum transferred by acoustic waves as they interact with and scatter off the object[1–5]. This force phenomenon has many practical applications and has been utilized for drug delivery[6], non-contact excitation[7–9], particle manipulation[10], and vibroacoustic spectrography of imaging acoustic responses of internal body tissues to mechanical excitation[11]. The acoustic radiation force is mathematically derived by integrating the acoustic radiation stress tensor over the object surface. The stress tensor is obtained by applying perturbations to the Navier-Stokes equations, which result in non-zero second order terms. The time-averaging method cancels out the linear, first-order term due to symmetrical oscillation, leading to a zero contribution over time. However, the nonlinear, second order terms do not cancel and produce the non-zero, time-averaged acoustic radiation force, which pushes the object toward the minimum potential[12–14]. Gorkov used the time-averaged potential and kinetic energies of an acoustic field, conceptualized as the gradient of a potential function, to determine the density, compressibility, and the geometry as functions of the acoustic radiation force. However, this formulation treats the object as an ideal, rigid particle and assumes it is acoustically small—i.e., its dimensions are much smaller than the wavelength of the incident acoustic field. As a result, Gorkov's theory fails to capture the more complex dynamical behaviour exhibited by acoustically large or deformable objects[15]. While several studies have

[1]Centre for Audio, Acoustics and Vibration, University of Technology Sydney, Sydney, Australia. [2z,2] Corresponding Authors: Mehdi.Akbarzadeh@uts.edu.au. ✉email: Mehdi.Akbarzadeh@uts.edu.au

investigated the acoustic wave-object interactions for object sizes larger than the wavelength limit[16–20], more research is required to understand their dynamics better.

Analytical representations, such as Gorkov's formulation, provide closed-form descriptions but are limited to acoustically small spherical objects; full numerical simulations extend to larger objects yet do not readily yield compact governing equations that reveal the underlying physical relationships. Sparse Identification of Nonlinear Dynamical (SINDy) is a data-driven algorithm that uses optimization to construct models by identifying a sparse set of candidate functions that describe the observed system behaviour[21–29]. For acoustically large objects, SINDy is applied here for the first time to evaluate its ability to extract explicit equations of motion directly from time-series data, while retaining predictive power and physical interpretability, which we are going to evaluate in this study.

We collect theoretical time series data for acoustically small spherical objects using the Gorkov formulation and compare this with experimental time series data generated from acoustically large objects and compare their bifurcation diagrams. In the acoustically small objects regime, we deliberately use synthetic data from Gorkov's formulation as an exact analytical reference to benchmark the accuracy of SINDy under controlled conditions. This ensures the method was validated before applying it to experimental data for large objects. Also, we limit our study to spherical objects to maintain consistency with Gorkov's formulation and to provide a clear baseline for validating the SINDy approach. This simplification allows direct benchmarking between theory, data, and experiment. Then we use a statistical analysis to identify the average quantities of coefficients used for SINDy to study acoustically large objects using the extracted equations with the aim to offer a framework to better understand the dynamics of arbitrary levitated objects, beyond the limitations of classical particle-based theories.

Methodology and extracting time series data

Here, we outline the process of generating and extracting theoretical and experimental time series data; additional details can be found in the Supplementary material (S1 and S2).

Extracting time series data using the Gorkov formulation

The Gorkov theory is valid for analysing acoustically small objects ($kr \ll 1$) with k and r representing the acoustic wave number and the object's radius[15]. By sinusoidally externally exciting an acoustic radiation field periodically using e.g. an electrodynamic shaker with amplitude of A_{ex}, and frequency of ω_{ex}, the levitated object vibrations at amplitude A_{o}, and frequency ω_{o}[30–36] and the resulting acoustic trap behaves as a Duffing-like oscillator, as follows[33]:

$$\ddot{\theta} + C_{\mathrm{D}} \left|\dot{\theta}\right| \dot{\theta} + C\left(\theta - \frac{\theta^3}{6}\right) - F\cos(\theta)\sin(\omega_{\mathrm{o}} t) = 0 \tag{1}$$

where $\theta = \gamma_2 z$ is the non-dimensional distance z of the object, from its equilibrium point. The remaining coefficients in Eq. (1) can be defined in terms of the properties of the object and the surrounding medium (details Supplementary material, S2). The nonlinearities considered in Eq. (1) originate from the acoustic potential field, which produces a Duffing-like stiffness response in the trap. Geometric nonlinearities, which typically arise from large-amplitude displacements or structural deflections, are not expected to be significant in Eq. (1) as the levitated sphere exhibited only small single-axis oscillations about its equilibrium point.

Extracting time series data using experimental set up

To measure the dynamics of an object trapped in an acoustic radiation force field ($kr > 1$) we setup an experiment using a Tiny-Lev levitator[33] (Fig. 1a). The levitator is driven by a square-wave having a peak-to-peak voltage of $9\,\mathrm{V}$ (function generator MFG-2260MFA), and is mounted vertically to an electrodynamic shaker (Bruel & Kjaer: LDS V201) driven by an amplifier (Bruel & Kjaer: LDS LPA100) using a sinusoidal signal of amplitude between 0 mm and $3\,\mathrm{mm}$ and a frequency range of 0 Hz to 250 Hz. The levitated object's vibration was measured using a tripod-mounted portable digital Laser Doppler Vibrometer (LDV, Polytec: PDV-100, sensitivity factor ($velo: 125\,\mathrm{mm/s/volt}$). The levitated object was measured through a $4\,\mathrm{mm}$ hole at the center of the levitator's head plate (Fig. 1b). The excitation and the LDV signal were monitored using an oscilloscope (Keysight: InfiniiVision DSOX2004a) recorded at $4\,\mathrm{kHz}$ sampling frequency. Protractors integrated into the tripod structures facilitated angular adjustments with a resolution of ± 0.5 ° (Fig. 1c).

Our experiments were conducted in a hemi-anechoic chamber at UTS, designed to minimize the effects of environmental noise and perturbations. The background noise was measured at the centre of the chamber following ISO 26,101 guidelines using a Brüel & Kjær microphone Type 4191 and a sound level meter Type 2250. The measurement includes both airborne and structure-borne noise and accounts for electrical noise in the instrumentation. The averaged background noise level over the one-third-octave bands cantered between 12.5 Hz and 20 kHz was found to be 17.1 dBA, which is sufficiently low to permit accurate and precise acoustic measurements. The noise floor measured with the Type 2250 is considered reliable, as it has much lower electrical self-noise compared to the microphone. The properties of the levitated object and the host fluid were assumed to be $\rho_0 = 1.19\ \frac{\mathrm{kg}}{\mathrm{m}^3}$ for the density of air, and for the speed of sound $c_0 = 340\ \frac{\mathrm{m}}{\mathrm{s}}$; while the drag coefficient was set to $C_{\mathrm{d}} = 1.06$[35,36]. We levitated the largest possible object—an Expanded Polystyrene (EPS) ball with a 1.87 mm radius—using the TinyLev setup at 40 kHz with a 9 V peak-to-peak voltage amplitude. This corresponds to $kr \approx 1.37$, well beyond the $kr \ll 1$ condition, placing the object outside the acoustically small-object regime where Gorkov's formulation applies with speed of sound of $c_1 = 570\ \frac{\mathrm{m}}{\mathrm{s}}$, and a density of $\rho_{\mathrm{p}} = 25\ \frac{\mathrm{kg}}{\mathrm{m}^3}$. The EPS ball was levitated in the middle between the two planes of the Tiny-Lev and its vibration

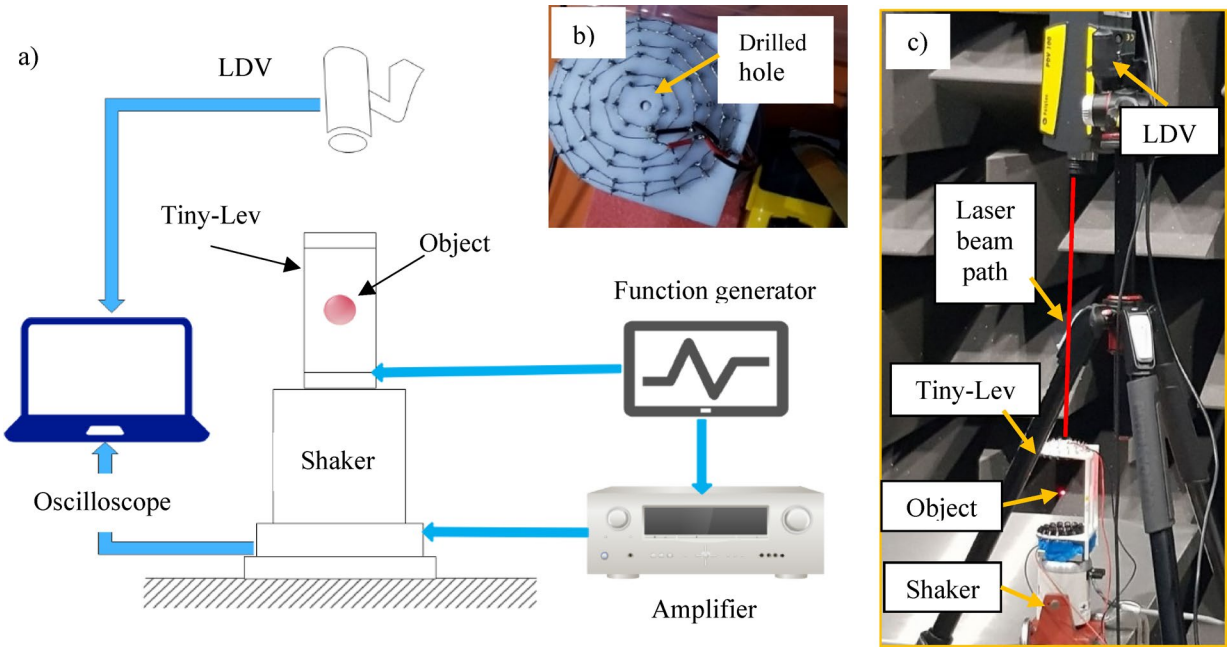


Fig. 1. Experimental setup in hemi-anechoic chamber. (**a**) Schematic and (**b**) drilled hole in the Tiny-Lev's top plate showing the object, and (**c**) Photograph of a levitated object within the Tiny-Lev mounted on the vibrating shaker, using an LDV for measurements.

velocity amplitude V_o had been estimated over voltage v_o, measured at the LDV and its sensitivity factor (*velo)* using

$$V_o = \frac{velo}{4} \times v_o \tag{2}$$

By assuming a sinusoidal motion its amplitude of displacement (A_o) is determined as

$$A_o = \frac{V_o}{\omega_o} \tag{3}$$

We carefully applied the GHKSS filter for noise reduction[37–40] to the extracted and converted time series to preserve the dynamics of the nonlinear system as much as possible (details Supplementary material, S3).

Results and discussion

We apply SINDy to analyse the dynamics of acoustically levitated objects. First, we validate its ability to reconstruct the original differential equation from time series generated by an analytical model based on Gorkov's formulation for acoustically small objects. We then extend the study to acoustically large objects, applying the analytical model and using SINDy to determine the governing equation. We implemented SINDy in MATLAB using the Thresholded Least Squares Algorithm (TLSA) as originally introduced by Brunton et al.[21]. This algorithm iteratively applies ordinary least squares (OLS) regression, followed by thresholding small coefficients to zero, thus promoting model sparsity. We used a fixed sparsity threshold of $\lambda = 10^{-6}$ across all experiments and simulations, which provided reliable results for the identification of governing equations. The choice of λ was based on trial-and-error tuning to achieve a balance between model accuracy and interpretability.

SINDy for acoustically small object: theoretical model

To benchmark SINDy for the theoretical study described in Eq. (1) and by using the Taylor series expansion for $\sin(\theta)$ up to order 3, we apply SINDy for an acoustically small object. Training data is collected over a time interval of two seconds, with a time step size of of $dt \approx \Delta t = 10^{-6}$ s to construct state space vectors as in the following

$$\frac{d}{dt}x_1 = x_2 = \dot{\theta}\,, \tag{4}$$

and

$$\frac{d}{dt}x_2 = \ddot{\theta} = -C_D\,|x_2|\,x_2 - Cx_1 + \frac{C}{6}{x_1}^3 + F\cos(x_1)\sin(\omega_o t) \tag{5}$$

to find the unknown coefficients $(C_{\mathrm{D}},\ C,\ F)$. A library of candidate functions is constructed, consisting of trigonometric and polynomial functions up to the fourth order using

$$\boldsymbol{\Theta}\ (\mathrm{X}) = \left[x_1,\ x_2, x_1x_2, \ldots, {x_1}^2, \ldots, {x_2}^4\right],$$

and

$$\mathbf{u}\left(\mathbf{X};t\right) = [x_1,\ x_2, \sin x_1 \sin x_2, \cos x_1, \cos x_2\ ]\ \sin\left(\omega_{\mathrm{o}} t\right) \tag{6}$$

Here, $\mathbf{X}\left(t\right) = [x_1, x_2]$ denotes the state vectors, and $\boldsymbol{\Theta}\ (\mathrm{X})$ is a library matrix of potential candidate functions to estimate the equation of motion, and $\mathbf{u}\left(\mathbf{X};t\right)$ is a library matrix of potential candidate functions to estimate external excitation part. The following parameters are used as properties of the object and the acoustic medium: the sound wave frequency of the acoustic field assumed to be $f = 40$ kHz, and the acoustic medium had a density of $\rho_0 = 1.19\ \frac{\mathrm{kg}}{\mathrm{m}^3}$, a speed of sound of $c_0 = 340\ \frac{\mathrm{m}}{\mathrm{s}}$, and a drag coefficient of $C_{\mathrm{d}} = 1.06$; the objects radius was $r = 0.71$ mm, and its material had a density of $\rho_1 = 25\ \frac{\mathrm{kg}}{\mathrm{m}^3}$ and speed of sound of $c_1 = 570\ \frac{\mathrm{m}}{\mathrm{s}}$. Using a fourth-order Runge-Kutta method on Eq. (1) we generate three benchmark systems of different dynamical responses to evaluate the performance of SINDy, using the amplitudes $\mathbf{A_{in}} = [0.15,\ 0.95,\ \text{and}\ 0.25]$ mm and excitation frequencies $\boldsymbol{\omega}_{\mathbf{o}} = [34,\ 10,\ 600]$ Hz based on the bifurcation diagram of the analytical solution[41]. Results are shown in Table 1.

According to Table 1, the comparison between the coefficients of the equation of motion obtained from the theoretical model and those derived using SINDy reveals a maximum difference of 0.04% for Benchmark (B) 1, 0.04% for B2, and 0.05% for B3. These studies indicates that SINDy is capable of accurately capturing and predicting the dynamics of the system based on the theoretical model. Table 1 shows that by constructing a library of candidate functions based on knowledge from the theoretical model, we ensure high accuracy[24–26]. For theoretical evaluation SINDy resilience to noise see Supplementary material, S4.

Application of SINDy to acoustically large object

We use time series data of three different experimental measurements to approximate the nonlinear equation of motion. Similar to the previous section we chose amplitudes, here $\mathbf{A_{in}} = [0.45{,}0.60,\ 0.70]$ mm, and but only a single excitation frequency of $\omega_{\mathrm{o}} = 32$ Hz and extracted 4 time series using the experimental setup depicted in Fig. 1. The following equations of motion are obtained after applying SINDy and averaging the extracted coefficients, with x representing the object's displacement from its static equilibrium point and $\dot{x}$ being the object's velocity:

$$\begin{aligned}
\begin{matrix} A_{\mathrm{in},1} \\ A_{\mathrm{in},2} \\ A_{\mathrm{in},3} \end{matrix} :=\ & \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \ddot{x} + \begin{bmatrix} 160{,}985 \\ 85{,}680 \\ 98{,}467 \end{bmatrix} x - \begin{bmatrix} 351.1 \\ 1{,}212.1 \\ 396.2 \end{bmatrix} \dot{x} - \begin{bmatrix} 667{,}030 \\ 358{,}700 \\ 565{,}600 \end{bmatrix} x^3 + \begin{bmatrix} 10{,}283 \\ 10{,}549 \\ 2{,}566 \end{bmatrix} x^2\dot{x} \\
& - \begin{bmatrix} 8.64 \\ 19.02 \\ 7.17 \end{bmatrix} x\dot{x}^2 + \begin{bmatrix} 0.2903 \\ 0.2470 \\ 0.0688 \end{bmatrix} \dot{x}^3 - \begin{bmatrix} 1{,}884 \\ 889 \\ 439 \end{bmatrix} x^3\dot{x} - \begin{bmatrix} 0.4985 \\ 5.2717 \\ 1.8023 \end{bmatrix} x^2\dot{x}^2 \\
& + \begin{bmatrix} 398{,}500 \\ 188{,}530 \\ 1{,}452{,}000 \end{bmatrix} x^5 + \begin{bmatrix} 60.452 \\ 170.34 \\ 34.93 \end{bmatrix} x^3\dot{x}^2 - \begin{bmatrix} 4.6975 \\ 1.6891 \\ 0.4018 \end{bmatrix} x^2\dot{x}^3
\end{aligned}$$

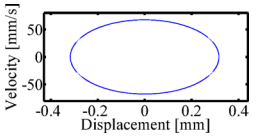
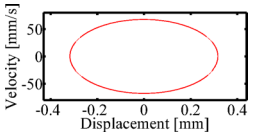
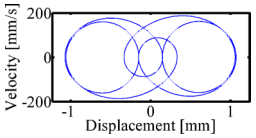
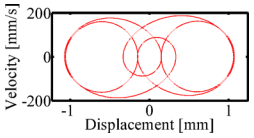
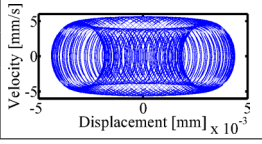
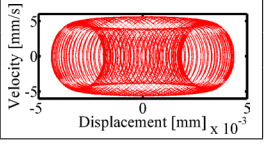
	Coeff.	Theory	SINDy	Phase space (Theory)	Phase space (SINDy)
B1	C_1	0.0149	0.0149		
	C_2	78,941	78,943		
	C_3	15,571	15,578		
B2	C_1	0.0149	0.0149		
	C_2	78,941	78,947		
	C_3	98,617	98,622		
B3	C_1	0.0149	0.0149		
	C_2	78,941	78,947		
	C_3	25,952	25,958		

Table 1. Application of SINDy to three theoretical benchmark (B) models based on the Gorkov formulation, including extracted coefficients (Coeff.) and phase portraits. Minimal deviations from the theoretical results are observed using an acoustically small object.

$$-\begin{bmatrix} 9255 \\ 14{,}647 \\ 17{,}641 \end{bmatrix} x\sin(\omega_o t) + \begin{bmatrix} 9{,}254 \\ 14{,}636 \\ 17{,}648 \end{bmatrix} \sin(x)\sin(\omega_o t) = 0 \tag{7}$$

The same extended function library, including polynomial terms in x and $\dot{x}$ up to fifth order, was used for both the small and large object cases. The differences in the final identified models are a result of the SINDy algorithm's sparsity-promoting structure. For the small object, most coefficients were eliminated, indicating relatively simpler dynamics. For the large object, more terms were retained which represent its richer nonlinear behaviour. This consistent approach avoids bias and allows the results to reflect the underlying system response. To generate a formulation using SINDy that not only displays the average of the measured data but also includes uncertainty bounds, we can use the mean value $\pm\ 2\sigma$ for each coefficient, which covers 95% of the data for a normal distribution. This approach provides a more conservative and comprehensive representation of uncertainty (see Tables S1-S3, in Supplementary material, S5). Figure 2 presents a statistical analysis of the variation in the calculated coefficients, using box plots[42–46] to illustrate the distribution of the SINDy-obtained coefficients for reconstructing the equation of motion based on the performed measurements.

To ensure comparability, all coefficients were normalized by dividing each by its respective maximum value. To view the original values of the coefficients used to reconstruct Eq. (7), refer to the coefficients listed in Tables S1, S2, and S3 in the Supplementary Material. Zero-valued coefficients are excluded from the plot to focus on the relevant dynamics. When a coefficient shows a small box plot in one subfigure and a large box plot in another—corresponding to different excitation amplitudes—it indicates that the coefficient's variability is sensitive to the excitation amplitude. The current uncertainty assessment, as depicted in Fig. 2, relies on statistical variation across repeated experimental runs and coefficient distributions. However, in the literature, there are other approaches to uncertainty quantification in sparse identification. One such approach, UQ-SINDy, is gaining traction[47], which integrates probabilistic modelling into the identification process by estimating distributions over both the model structure and coefficients[47]. UQ-SINDy has the potential to yield confidence intervals directly on terms within the governing equation and support model selection under uncertainty. Although beyond the scope of this study, incorporating uncertainty-aware methods such as UQ-SINDy could enhance model reliability in future work, particularly under noisy or ambiguous experimental conditions.

When the governing equation of motion, Eq. (7), changes only in its coefficients due to variations in the bifurcation parameter, A_{in}, we can describe this as a parametric variation within the same dynamical model. Figure 3 compares the measured experimental data, the filtered time series, and the SINDy results, along with the power spectral density (PSD) of the LDV measured signal before and after filtering, as well as the PSD according to the SINDy results in Eq. (7) for three different measurements by changing the amplitude of the external excitation, $\mathbf{A_{in}}$, respectively.

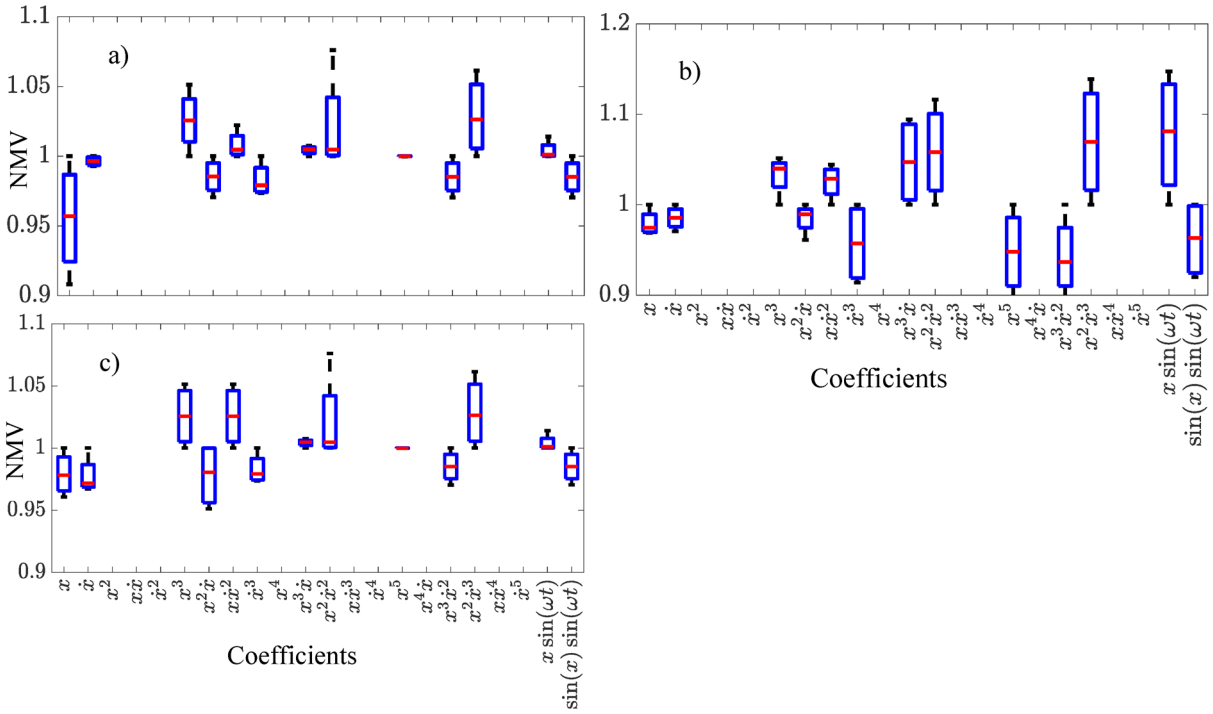


Fig. 2. (**a**) Box plot analysis of the variation in coefficients derived using the SINDy method across four measurements. Zero-valued coefficients are excluded from the plot. The analysis highlights the statistical distribution and variability of normalized measured values (NVM) used to reconstruct the nonlinear equations of motion: (**a**) Eq. (7), (**b**) Eq. (8), and (**c**) Eq. (9)).

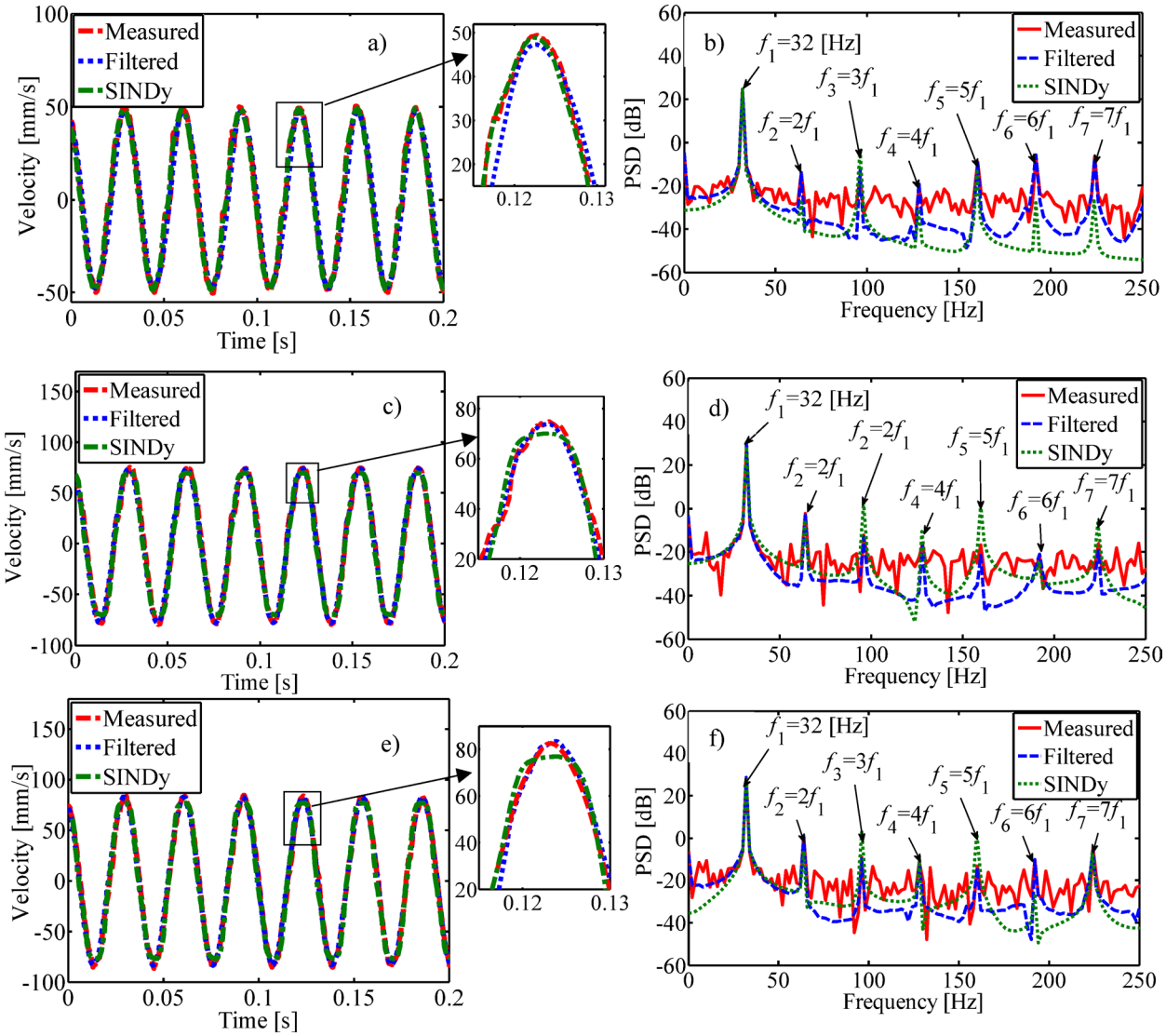


Fig. 3. Comparison of experimentally measured data and SINDy predictions using averaged coefficients with filtered data and PSD of the original signal measured by LDV before applying a filter, after applying the filter for: (**a**, **b**) $A_{\mathrm{in},1} = 0.60$ mm, (**c**, **d**) $A_{\mathrm{in},2} = 0.45$ mm, (**e**, **f**) $A_{\mathrm{in},3} = 0.70$ mm. All experimental measurements were performed at the fixed frequency $\omega_{\mathrm{o}} = 32$ Hz. These plots highlight the accuracy of SINDy in predicting the system dynamics.

The time series presented in Fig. 3(a–f) correspond to steady-state oscillations of the levitated object, recorded after initial transients had subsided under continuous acoustic excitation. These intervals were deliberately selected for modelling, as they exhibit consistent and repeatable dynamics representative of the system's nonlinear behaviour. Transient segments—such as those arising during start-up or excitation changes—were excluded to avoid variability that does not reflect the system's long-term response. By focusing on well-defined attractor states, the identified governing equations capture the essential features of the sustained dynamics and remain robust within the experimentally accessible range.

In Fig. 3, the alignment of the PSD peaks from the measured signal, filtered signal and SINDy-reconstructed data suggests that SINDy successfully captured the essential dynamics of the system. It shows the fundamental frequency of 32 Hz and its integer multiples of a fundamental frequency, and its harmonics $2f, 3f, 4f$, indicative for a nonlinear period-1 limit cycle being present[41,42]. The PSD further confirms that the approximation provided is quite accurate. Since SINDy relies on a predefined library of candidate functions whose suitability affects the model's accuracy, the library we had chosen has been appropriate for the problem at hand[24–26].

To further validate the results, we next apply the Gorkov formulation and expand the Taylor series to the fifth order to study small objects levitated in air subjected to external excitation (see Supplementary Material S2). This test will provide insights into whether the equations extracted from SINDy will be structurally different from the original Gorkov formulation. The resulting governing equation of motion is therefore derived as follows:

$$\ddot{\theta} + C_1\theta + C_2\theta^3 + C_3\theta^5 + C_D\dot{\theta}\left|\dot{\theta}\right| = F\cos(\theta)\sin(\omega_o t) \tag{8}$$

where the coefficients C_i with $i = 1,2,3$ related to the object and fluid properties, C_D related to viscosity, and F represents the amplitude of the external excitation. By using SINDy on a large spherical object, Eq. (8) takes the following form

$$\ddot{\theta} + c_1\theta + c_2\theta^3 + c_3\theta^5 + \mathcal{F}(\theta, \dot{\theta}) = (f_1\theta + f_2\sin(\theta))\sin(\omega_o t) \tag{9}$$

when $\mathcal{F}(\theta, \dot{\theta})$ is a nonlinear function and can be written as

$$\mathcal{F}(\theta, \dot{\theta}) = d_1\dot{\theta} + d_2\theta^2\dot{\theta} + d_3\theta\dot{\theta}^2 + d_4\dot{\theta}^3 + d_5\theta^3\dot{\theta} + d_6\theta^2\dot{\theta}^2 + d_7\theta^3\dot{\theta}^2 + d_8\theta^2\dot{\theta}^3 \tag{10}$$

in which all c_i coefficient in Eq. (9) and d_i coefficient in Eq. (10) are related to the external excitation amplitude, and $\mathcal{F}(\theta, \dot{\theta})$ is a nonlinear function of θ, $\dot{\theta}$ that can be replaced instead of $C_D\dot{\theta}\left|\dot{\theta}\right|$ in Eq. (10). Also, the right-hand side of Eq. (10), $F\cos(\theta)\sin(\omega_o t)$, has been replaced by $(f_1\theta + f_2\sin(\theta))\sin(\omega_o t)$. All coefficients in Eqs. (9) and (10) were obtained using SINDy. Because the governing equations were identified directly from LDV-measured velocity time series, SINDy inherently incorporates hidden influences such as temperature variations, medium viscosity changes, and other secondary effects into the fitted coefficients. In this study, we did not isolate these effects in an independent analysis; instead, their impact is implicitly embedded in the reconstructed dynamics. A notable feature here is the presence of mixed nonlinear terms, which are not captured in classical formulations. All coefficients in Eqs. (9) and (10) admit a direct interpretation: the linear displacement coefficient sets the effective trap stiffness, odd displacement terms (e.g., cubic) produce amplitude-dependent hardening/softening, linear and higher-order velocity terms quantify dissipative losses, and mixed displacement–velocity terms indicate nonlinear coupling between restoring and dissipative dynamics. In this way, the reconstructed equation not only reproduces the measured responses but also highlights additional physical effects that warrant further investigation. From measurements one to three, we find that f_1 and f_2 are very close to each other and can be treated as a single coefficient f. The ratio of f to A_{in} in all three equations is about 25,000. This suggests a consistent relationship between f and A_{in}, which might be an important characteristic of the system. In comparison with Eq. (8), Eq. (9) indicates a stronger influence of viscosity and increasingly important interaction-effects between object, fluid, and the system's dynamic response. The SINDy method reveals important nonlinear effects in acoustically levitated objects that classical models like Gorkov's miss. Specifically, cubic and mixed nonlinear terms show complex interactions between acoustic forces, drag, and inertia at larger object sizes. These nonlinear terms depend on excitation amplitude, meaning the trap's stiffness and damping change with vibration strength. Also, strong velocity-dependent terms for large objects indicate that energy loss grows at higher excitations—something hard to capture with traditional analysis. This new understanding of scaling and nonlinear behaviour can help improve control and characterization in future levitation systems.

Figure 4a presents the bifurcation diagram for an acoustically large object at $k = 1$ and $r = 1.87$ mm, where the Gorkov formulation (red line) fails to predict the experimental system behaviour (blue dots). Figure 4b uses instead of Gorkov the extracted equation from SINDy (red line) as analytical result, with experimental data showing up again as blue dots, which shows a much better match.

Figure 4a highlights the divergence between the theoretical bifurcation structure and the experimental results at higher excitation amplitudes. Specifically, Gorkov's model fails to capture the multi-branch bifurcation patterns observed in the experiment, indicating its inability to represent the system's nonlinear transitions and amplitude-dependent behaviour. In contrast, Fig. 4b demonstrates that the SINDy-derived model captures both the fundamental and higher-order nonlinearities, providing significantly better agreement with the experimental bifurcation data. The results obtained using statistical analysis are presented in Fig. 4c. The red markers represent the lower bond predicted by the SINDy model, while the black dots illustrate the statistical upper uncertainty bounds also the experimental data are also included for comparison. The bifurcation diagrams in Fig. 4 reflect how the system responds to changes in the excitation amplitude. Although we do not derive a closed-form normal form using analytical methods such as centre-manifold reduction, the SINDy-identified equations can be considered a data-driven normal form, where the external excitation amplitude acts as a bifurcation parameter. These results suggest that SINDy provides not only an interpretable model but also a practical tool for reconstructing nonlinear bifurcation behaviour in experimentally limited regimes. In our previous study[33], we demonstrated that acoustically small spherical levitated objects exhibit harmonic oscillations under external excitation, with nonlinear behaviours such as Duffing-like responses and bifurcations. Building on this, the current study uses a data-driven approach to reconstruct the governing nonlinear equations from time series data, capturing amplitude-dependent and higher-order effects. Together, these studies provide complementary insights into the dynamics of levitated objects, ranging from small spheres under controlled excitation to large objects that exceed the limits of classical formulations, typically bound by the Rayleigh limit.

Conclusions

In this study, we applied the Sparse Identification of Nonlinear Dynamical Systems (SINDy) to reconstruct a spherical levitated object's nonlinear differential equation of motion when trapped in an externally excited acoustic radiation force field, using theoretical and experimental time series. This study demonstrates a two-stage application of SINDy: first, as a validation tool using synthetic data extracted from Eq. (1) for the acoustically small object regime, and second, as a discovery tool applied to experimental data for acoustically large objects.

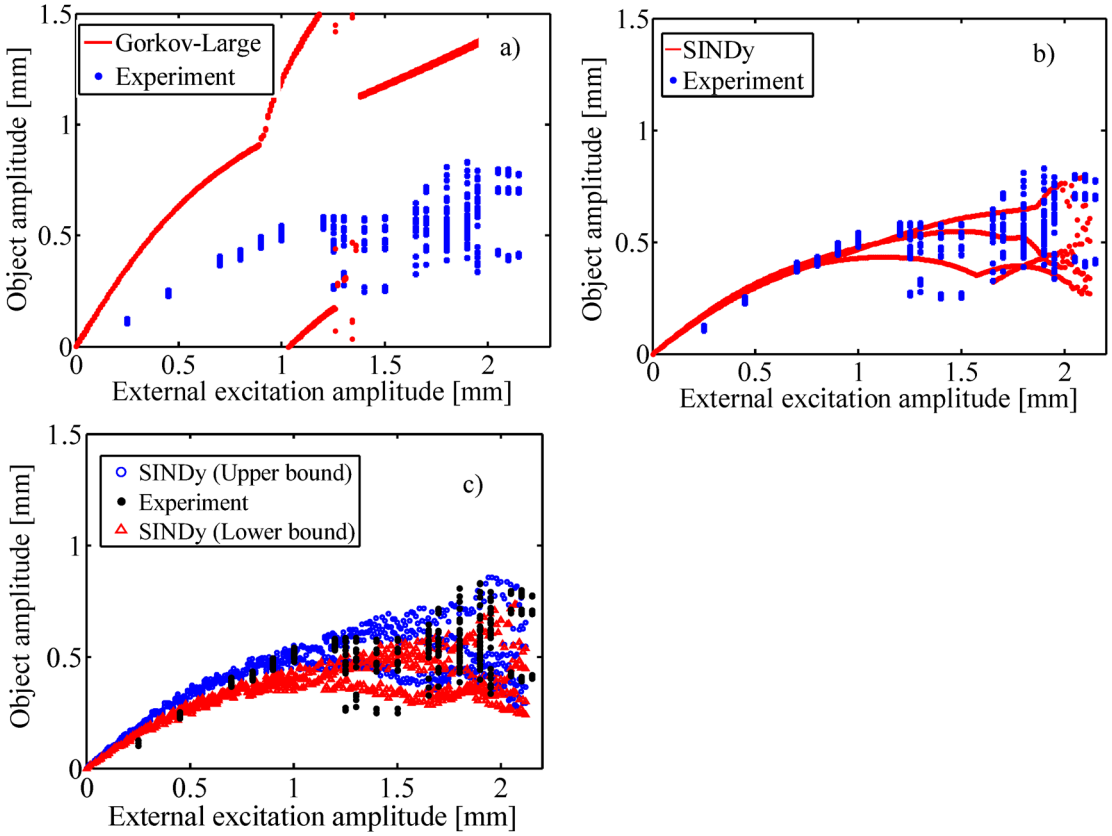


Fig. 4. (**a**) Bifurcation diagram for an acoustically large object: analytical (solid line) vs. experimental data (dots), showing discrepancies due to Gorkov's small-object assumption. (**b**) Comparison of the SINDy results (red), and experimental data (blue), highlighting deviations at higher excitation amplitudes due to the polynomial-based library limitations. (**c**) Comparison of SINDy-derived results with experimental data. The black dots represent the upper uncertainty bond predictions from SINDy, while the red triangles indicate the lower uncertainty bounds in comparison with experimental measurements.

We found that for the acoustically small object regime, the extracted coefficients differed by less than 0.05% from the exact analytical model. The experimental data used for the second stage corresponds to the system's response under a period-1 oscillatory regime, which was the only stable and repeatable dynamic state achievable within the limits of our acoustic levitation setup. While this restricts experimental validation to a single regime, it still provides a rich basis for extracting the governing dynamics. Importantly, the structure of the identified model is not hand-tuned but emerges purely from the SINDy algorithm's sparse regression process, using the same candidate library functions across both acoustically small and large object cases. This allows for a direct comparison: in the large-object regime, SINDy revealed significantly higher-order and mixed nonlinear terms that were absent in the small-object case, highlighting how the model captures physically meaningful transitions beyond the assumptions of classical formulations. Here, SINDy employed as a data-driven framework to bridge the gap between theoretical limits and experimental observations of nonlinear object dynamics, demonstrates that, even within a limited dynamic regime, the identified equations carry clear physical signatures and offer insights into amplitude-dependent dynamics not previously characterized in literature.

We used box plots to visualize distributions and to assist in identifying key coefficients in SINDy models, to reveal their sensitivity to changes in external excitation. For acoustically large objects, SINDy reveals nonlinear dependencies between the external excitation amplitude and the extracted coefficients in the governing equation. These relationships indicate that the system's response is strongly amplitude-dependent. Although the chosen library may not be used to fully capture all aspects of the bifurcation behaviour, it provides valuable insights into the dominant dynamics of the system. The three datasets in Fig. 3 were restricted to period-1 responses at a fixed frequency, since only under these conditions could stable and interpretable governing equations be reliably extracted using SINDy. In nonlinear systems, a stable limit cycle is not just a simple periodic solution but also encodes the essential nonlinearities of the underlying dynamics, making it a suitable basis for model extraction. Although additional validation under different amplitudes or transient conditions would be valuable, this initial demonstration confirms that the reconstructed equations capture the essential dynamics, with systematic trends and bifurcation behaviour (Fig. 4) further supporting their generality. Beyond reproducing the measured responses, the SINDy coefficients serve as indicators of the balance between stiffness, damping, and nonlinear coupling, which shifts with excitation amplitude. This shows that mechanisms such as viscous losses, acoustic streaming, and secondary radiation leave measurable signatures in the governing equations. Applying the same

approach to different object sizes and geometries can reveal how these mechanisms scale in nonlinear acoustic trapping.

Also, future research could apply SINDy to experimental measurements in the acoustically small-object regime to directly compare with Gorkov's formulation. Although our study was restricted to geometrically spherical objects, it is worth noting that SINDy itself imposes no restrictions on object geometry and can be used for non-spherical objects, too, which could be an expansion of our current study. Further, SINDy could be applied to experimental measurements of also larger objects, using different levitator devices, to quantify more specifically the functional relationships of coefficients and nonlinear terms. This could provide a basis for using levitation as a more comprehensive dynamic testing platform, where variations in SINDy-derived coefficients can reveal material properties that cannot be measured directly using conventional methods.

Data availability

The datasets generated and analysed during the current study are available in the Zenodo repository: https://zenodo.org/records/16033764.The data include nonlinear time series measurements from the acoustic levitation experiments described in the study, "Sparse identification of nonlinear dynamics applied to the levitation of acoustically large objects." While the full dataset is provided, only the first three Period-1 oscillation cycles were used in the analysis presented in the paper. The accompanying MATLAB code for data processing and system identification was written and tested using MATLAB R2023a. For further inquiries, please contact the corresponding author.

Received: 13 July 2025; Accepted: 14 October 2025
Published online: 19 November 2025

References

1. Torr, G. R. The acoustic radiation force. *Am. J. Phys.* **52** (5), 402–408 (1984).
2. Silva, G. T., Chen, S., Greenleaf, J. F. & Fatemi, M. Dynamic ultrasound radiation force in fluids. *Phys. Rev. E Stat. Nonlin Soft Matter Phys.* **71** (5), 056617–056617 (2005).
3. Bruus, H. Acoustofluidics 7: the acoustic radiation force on small particles. *Lab. Chip*. **12** (6), 1014–1021 (2012).
4. Zhang, P., Bachman, H., Ozcelik, A. & Huang, T. J. Acoustic microfluidics. *Annu. Rev. Anal. Chem.* **13** (1), 17–43 (2020).
5. Kandemir, M., Beelen, M., Wagterveld, R., Yntema, D. & Keesman, K. Dynamic acoustic fields for size selective particle separation on centimeter scale. *J. Sound Vib.* **490** (6), 115723 (2021).
6. Mabvaro, T. & Browne, J. Characterisation of the acoustic output of three sonoporation drug delivery ultrasound systems using an acoustic radiation force balance. *Phys. Med.* **28** (4), 343 (2012).
7. Huber, T. M., Abell, D. C., Mellema, C. B., Spletzer, M. & Raman, A. Mode-selective noncontact excitation of microcantilevers and microcantilever arrays in air using the ultrasound radiation force. *Appl. Phys. Lett.* **97**, 21410 (2010).
8. Huber, T. M., Beaver, J. R. & Helps, N. M. Noncontact modal excitation of a classical guitar using ultrasound radiation force. *Exp. Tech.* **37**, 38–46 (2013).
9. Chen, S., Sabato, A., Niezrecki, C., Avitabile, P. & Huber, T. Characterization and modeling of the acoustic field generated by a curved ultrasound transducer for non-contact structural excitation. *J. Sound Vib.* **432**, 33–49 (2018).
10. Nakahara, J. & Smith, J. R. Acoustic balance: weighing in ultrasonic non-contact manipulators. *IEEE Robot Autom. Lett.* **7** (4), 9145–9150 (2022).
11. Rifu, K., Watanabe, J., Sasanuma, H. & Taniguchi, N. Evaluation of the elasticity of the pancreas using acoustic radiation force impulse elastography in patients with acute pancreatitis: A systematic review and meta-analysis, ultrasound med. *Biol* **48** (3), 406–413 (2022).
12. Chu, B. T. & Apfel, R. E. Acoustic radiation pressure produced by a beam of sound. *J. Acoust. Soc. Am.* **72** (6), 1673–1687 (1982).
13. Mase, G. T. & Mase, G. E. Continuum mechanics for engineers, Second Edition, CRC Press LLC, (1999).
14. Bruus, H. Acoustofluidics 2: perturbation theory and ultrasound resonance modes. *Lab. Chip*. **12** (1), 20–28 (2012).
15. Gorkov, L. P. On the forces acting on a small particle in an acoustical field in an ideal fluid. *Sov Dokl. Phys.* **6**, 773–775 (1962).
16. King, L. V. On the acoustic radiation pressure on spheres. *Proc. Royal Soc. Lond. A*. **147** (861), 212–240 (1934).
17. Yosioka, K. & Kawasima, Y. Acoustic radiation pressure on a compressible sphere. *Acustica* **5** (3), 167–173 (1955).
18. Hasegawa, T., Saka, K., Inoue, N. & Matsuzawa, K. Acoustic radiation force experienced by a solid cylinder in a plane progressive sound field. *J. Acoust. Soc. Am.* **83** (5), 1770–1775 (1988).
19. Doinikov, A. A. Acoustic radiation forces classical theory and recent advances. *Transw. Res. Netw. Chap*. **3**, 39–67 (2003).
20. Mitri, F. G. Theoretical calculation of the acoustic radiation force acting on elastic and viscoelastic cylinders placed in a plane standing or quasistanding wave field. *Eur. Phys. J. B*. **44**, 71–78 (2005).
21. Brunton, S. L., Proctor, J. L. & Kutz, J. N. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proc. Natl. Acad. Sci.* **113** (15), 3932–3937 (2016).
22. Brunton, S. L., Proctor, J. L. & Kutz, J. N. Sparse identification of nonlinear dynamics with control (SINDYc). *IFAC-PapersOnLine* **49** (18), 710–715 (2016).
23. Stender, M., Oberst, S. & Hoffmann, N. P. Recovery of differential equations from impulse response time series data for model identification and feature extraction. *Vibration* **2** (1), 25–46 (2019).
24. Wentz, J., Doostan, A. & Hoffmann, N. P. Derivative-based SINDy (DSINDy): addressing the challenge of discovering governing equations from noisy data. *Comput. Methods Appl. Mech. Eng.* **413**, 116096 (2023).
25. Champion, K., Brunton, S. L. & Kutz, J. N. Data-driven discovery of coordinates and governing equations. *PNAS* **116** (45), 22445–22451 (2019).
26. Ugolini, A. R., Breschi, V., Manzoni, A. & Tanelli, M. SINDy vs hard nonlinearities and hidden dynamics: a benchmarking study. *IFAC-Pap* **58** (15), 49–54 (2024).
27. Camps-Valls, G. et al. Discovering causal relations and equations from data. *Phys. Rep.* **1044**, 1–68 (2023).
28. Bishnu, S. K., Alnouri, S. Y. & Al-Mohannadi, D. M. Computational applications using data driven modeling in process systems: A review. *Digit. Chem. Eng.* **8**, 100111 (2023).
29. Abdullah, F. & Christofides, P. D. Data-based modeling and control of nonlinear process systems using sparse identification: an overview of recent results. *Comput. Chem. Eng.* **174**, 108247 (2023).
30. Dolev, A., Davis, S. & Bucher, I. Noncontact dynamic oscillations of acoustically levitated particles by parametric excitation. *Phys. Rev. Appl.* **12**, 034031 (2019).
31. Lu, X. et al. Dynamic acoustic levitator based on subwavelength aperture control. *Adv. Sci.* **8**, 2100888 (2021).
32. Hasegaw, K. & Murata, M. Oscillation dynamics of multiple water droplets levitated in an acoustic field. *Micromachines* **13**, 1373 (2022).

33. Akbarzadeh, M., Oberst, S. & Halkon, B. Manipulation of an acoustically levitated object using externally excited standing waves. *J. Acoust. Soc. Am.* **157** (3), 1852–1861 (2025).
34. Andrade, M. A. B., Ramos, T. S., Okina, F. T. A. & Julio, C. A. Nonlinear characterization of a single-axis acoustic levitator. *Rev. Sci. Instrum.* **85**, 045125 (2014).
35. Fushimi, T., Hill, T. L., Marzo, A. & Drinkwater, B. W. Nonlinear trapping stiffness of mid-air single-axis acoustic levitators. *Appl. Phys. Lett.* **113** (3), 34102 (2018).
36. Goossens, W. R. A. Review of the empirical correlations for the drag coefficient of rigid spheres. *Powder Technol.* **352**, 350–359 (2019).
37. Grassberger, P., Hegger, R., Kantz, H., Schaffrath, C. & Schreiber, T. On noise reduction methods for chaotic data. *Chaos* **3**, 127 (1993).
38. Urbanowicz, K. & Kantz, H. Improvement of speech recognition by nonlinear noise reduction. *Chaos* **17** (2), 023121063114 (2007).
39. Moore, J. M., Small, M. & Karrech, A. Improvements to local projective noise reduction through higher order and multiscale refinements. *Chaos* **25** (6), 063114 (2015).
40. Oberst, S., Marburg, S. & Hoffmann, N. Determining periodic orbits via nonlinear filtering and recurrence spectra in the presence of noise. *Procedia Eng.* **199**, 772–777 (2017).
41. Akbarzadeh, M., Oberst, S., Sepehrirahnama, S. & Halkon, B. Sensitivity and bifurcation analysis of an analytical model of a trapped object in an externally excited acoustic radiation force field, Conf. Proceedings of NOVEM23. Auckland, 1–10.
42. Schuster, H. G. & Just, W. Deterministic Chaos: An Introduction, 2nd ed., Wiley-VCH, edition. (2005).
43. Garland, J. & Bradley, E. Prediction in projection. *Chaos* **25** (12), 123108 (2015).
44. Rousseeuw, P. J., Ruts, I. & Tukey, J. W. The bagplot: A bivariate boxplot. *Am. Stat.* **53** (4), 382–387 (1999).
45. Williamson, D. F., Parker, R. A. & Kendrick, J. S. The box plot: A simple visual method to interpret data. *Ann. Intern. Med.* **10** (11), 916–921 (1989).
46. Papoulis, A., Pillai, S. U., Probability, R., Variables & Processes, S. 4th ed., McGraw-Hill, (2002).
47. Hirsh, S. M., Barajas-Solano, D. A. & Kutz, J. N. Sparsifying priors for bayesian uncertainty quantification in model discovery. *R Soc. Open. Sci.* **9**, 211823 (2022).

Acknowledgements

The authors would like to thank Dr. Shahrokh Sepehrirahnama for some early discussions with the lead author.

Author contributions

Mehdi Akbarzadeh: Conceptualisation, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing – Review and Editing, Visualisation. Benjamin Halkon: Investigation, Writing - Review and Editing, Supervision. Sebastian Oberst: Conceptualisation, Methodology, Investigation, Validation, Resources, Writing - Review and Editing, Supervision, Project administration, Funding acquisition.

Funding

The first author acknowledges a UTS President's Scholarship. This research is partially supported by the Australian Research Council Discovery Project DP240101536.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-24666-5.

Correspondence and requests for materials should be addressed to M.A.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025